Analysis of torsional vibration in internal combustion engines: modelling and experimental validation

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Abstract: This paper reports on a study of the crankshaft torsional vibration phenomenon in internal combustion engines. The steady state of the state equation is solved by the transition state matrix and the convolution integral. This formulation is applied to the model of a six-cylinder diesel engine manufactured by MWM International Motores®. The analyses consider a rubber and viscous damper assembled to the crankshaft front-end. An analysis of the torsional vibrations indicates the dynamic loading on each crankshaft section, and these loads can be applied as boundary conditions in a finite element model to predict the safety factor of the component and to compare the system's behaviour with rubber and viscous damper options. This study highlights the importance of torsional vibration analyses in the structural dimensioning of crankshafts. The results of the torsional vibration amplitude are compared with measured values to experimentally validate the proposed mathematical model.

Keywords: torsional vibrations, internal combustion engines, viscous dampers, rubber dampers

1 INTRODUCTION

A crankshaft is subjected to periodic dynamic loads, generating vibrations and stresses that must be quantified to ensure the structural integrity of the component. Today, due to technical, commercial and environmental requirements, internal combustion engines (ICEs) must operate with high cylinder pressures and the components must be optimized for the best performance.

Modern calculation methods allow for the precise determination of stress levels in the crankshaft's critical regions, as well as evaluation of the fatigue strength. Thus, it is possible to consider design margins that ensure sufficient reliability to avoid structural failures and oversizing of the components.

This study began with an analysis considering no torsional vibration damper (TVD) to adjust and calibrate the engine's internal damping and to check the natural frequencies of the system. The second step of the study involved a rubber absorber, whose power dissipation capability was checked for structural integrity. Finally, calculations were performed considering a viscous damper in the system (Fig. 1).

Complete torsional vibration analysis (TVA), including calculations of the vibration amplitudes at the crankshaft front-end, actuating dynamic torques in rear and front connections, damper power dissipation, and rubber shear stress, will be performed for the aforementioned cases.

Crankshaft torsional vibrations occur in ICEs due to the periodic nature of the actuating torque. Basically, the TVA performed here began by outlining a mathematical model to represent the system's dynamic characteristics, such as inertias, torsional stiffness, and damping. The excitation torque was then calculated considering the gas load and inertia forces of the moving parts and a Fourier series expansion of this torque was performed. The harmonics thus obtained were applied to the corresponding crank throws, considering the ignition time of the engine.

The technical features of the engine (Fig. 2) under analysis are listed below:

(a) firing order: 1-5-3-6-2-4;
(b) four-stroke cycle;

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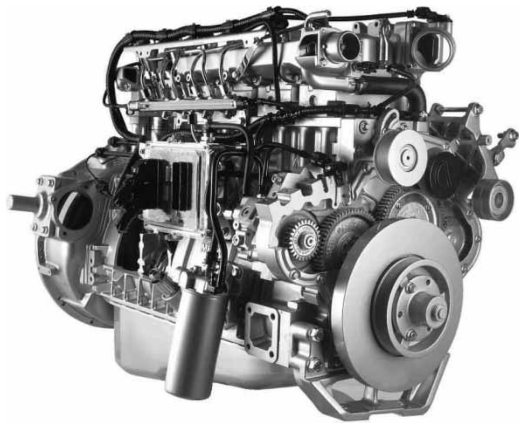

Fig. 1 The 7.2 liter diesel engine

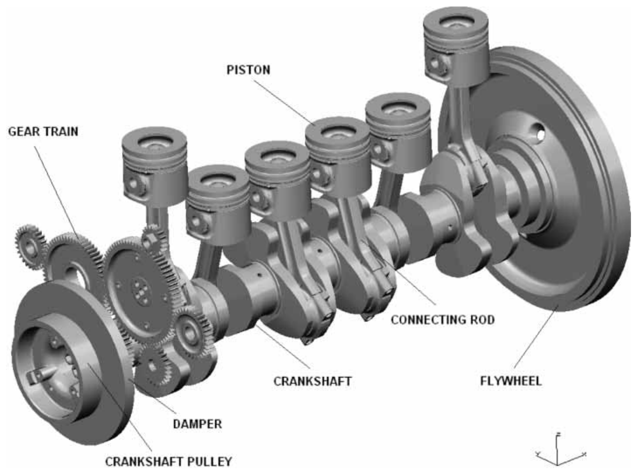

Fig. 2 Crankshaft system considered in the TVA

(c) connecting rod length: 207 mm;
(d) cylinder bore: 105 mm;
(e) piston stroke: 137 mm;
(f) oscillating masses: 2.521 kg;
(g) maximum torque: 1100 Nm at 1200 r/min;
(h) maximum power: 228 kW at 2200 r/min;
(i) maximum engine speed: 2550 r/min.

2 LITERATURE REVIEW

Some systems can present excessive torsional vibrations at specific engine speeds. Draminsky [**1**] was one of the first researchers who studied these phenomena. Hestermann and Stone [**2**] concluded that these unexpected large angular displacements in multiples of the engine speed occur due to the variable inertia characteristics of the crank mechanism.

In the past, the effects of variable inertia of ICEs were considered negligible and were not included in calculations. Recently, these secondary effects were verified and checked and were found to be responsible for many structural failures of crankshafts. Pasricha [**3**] included these effects to Draminsky's earlier studies and concluded that, in some cases, the interaction of these secondary forces can be extremely dangerous for crankshafts.

Other researchers such as Johnston and Shusto [**4**] developed and applied a technique to predict the behaviour of torsional vibrations in ICEs in the transient and steady-state response by the modal superposing method. The results of this analysis were compared with experimental values for the validation of a mathematical model.

Brusa *et al.* [**5**] studied the introduction of functions taking into account the variation of inertia in the crankshaft's angular position and the coupling of axial-flexural vibrations. These considerations substantially increased the number of equations to be solved and the computational cost, but the final results were more accurate for the cases reported in their article. Song *et al.* [**6**], who analysed the coupling effect of torsional and axial vibration in the crankshafts, concluded that large displacements are expected when the natural axial and torsional frequencies are equal, or when the former is two-fold greater than the latter.

The torsional damping coefficients of ICEs were initially estimated by researchers Hartog [**7**] and Wilson [**8**]. These parameters were obtained from empirical determinations and, in most cases, were inaccurate, generating considerable variations in the dynamic response of the analysed systems. Theoretical and hybrid models to estimate damping coefficients were proposed by Iwamoto and Wakabayashi [**9**], who considered analytical relations between the damping and other measurable engine parameters.

Wang and Lim [**10**] accurately estimated the absolute damping of a single-cylinder engine powered by an electric motor. The first two mode shapes of the system were considered and the absolute damping coefficients were obtained as a function of the crank angle. Many researchers consider absolute damping a constant at all engine speeds and in every crankshaft position.

In a study on the torsional vibrations of a six-cylinder diesel engine, Honda and Saito [**11**] attempted to reduce the vibratory effects with a rubber TVD. They used the transition state matrix methodology and found that the torsional stiffness of the rubber damper played a more significant role in the system's characteristics than the engine's internal damping and even TVD damping. This stiffness is determined mainly by the geometry and chemical composition of the rubber.

The excitation torque is usually considered constant and equal in all cylinders. This holds true only for new engines and considerable variations in the shape of the cylinder's internal pressure curves can be expected during the engine's operational life. Maragonis [**12**] studied the variation of the excitation load through the cylinders due to the wear of piston rings and liner and reported some interesting results.

3 THEORETICAL MODELLING

Crankshafts are subject to torsional, axial, and flexural vibrations due to the periodic nature of excitation loading. In this study, only torsional vibrations were analysed, requiring the determination of an equivalent mathematical model of the system.

An analysis was made considering a viscous TVD assembled in the crankshaft. Another analysis considered a double mass rubber damper to reduce the torsional amplitudes. Figure 3 illustrates the model for a single mass viscous damper, while Fig. 4 presents the model for a double mass rubber damper analysis.

3.1 Inertias

The inertias of the system, such as flywheel, pulleys, crank throws, and TVD rings, can be determined using

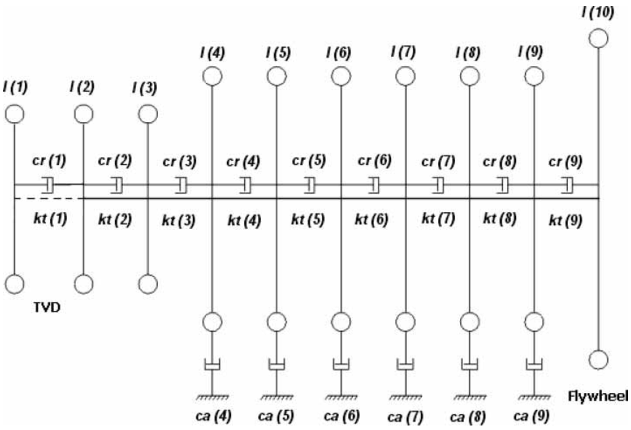

Fig. 3 Equivalent model considering a single mass viscous TVD

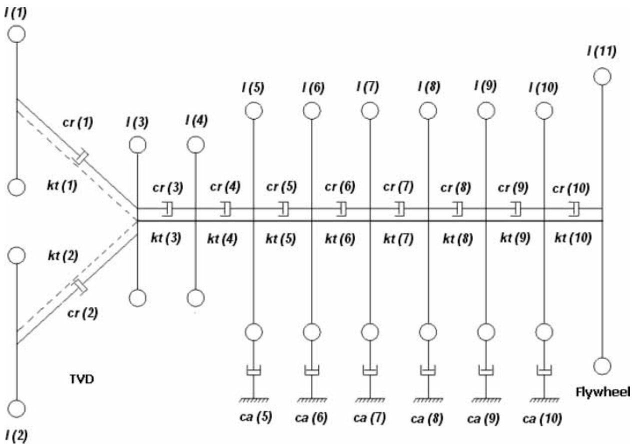

Fig. 4 Equivalent model considering a double mass rubber TVD

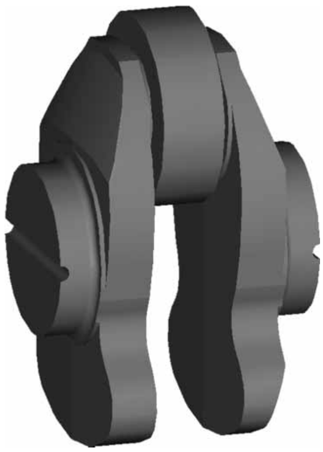

Fig. 5 Geometric model for calculating inertia

CAD software. The inertia of a single crank throw is calculated based on the model shown in Fig. 5.

The connecting rod mass was divided into two masses. One of them had a purely rotating movement, 'm_{rb}', while the other had an oscillating movement, 'm_{ab}'. Figure 5 indicates that the rotating mass of the con rod was considered in the calculation of crank throw inertia.

The con rod mass 'm_b', including bolts, bearings, and bushing, can be divided according to the following methodology (Fig. 6)

$$m_{ab} = \frac{m_b L_2}{L} \tag{1}$$

$$m_{rb} = \frac{m_b L_1}{L} \tag{2}$$

Engines usually have a gear train for power transmission to other devices. The inertia of this system is considered in the equivalent model. For example, the equivalent inertia of a device driven by gear 2 with a rotational speed n_2 related to gear 1 with a rotational

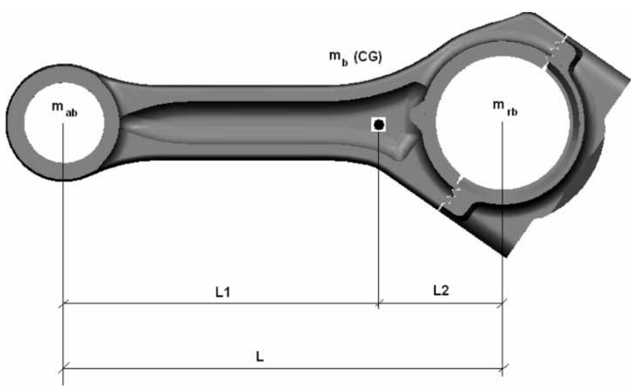

Fig. 6 Dimensions considered for the division of con rod masses

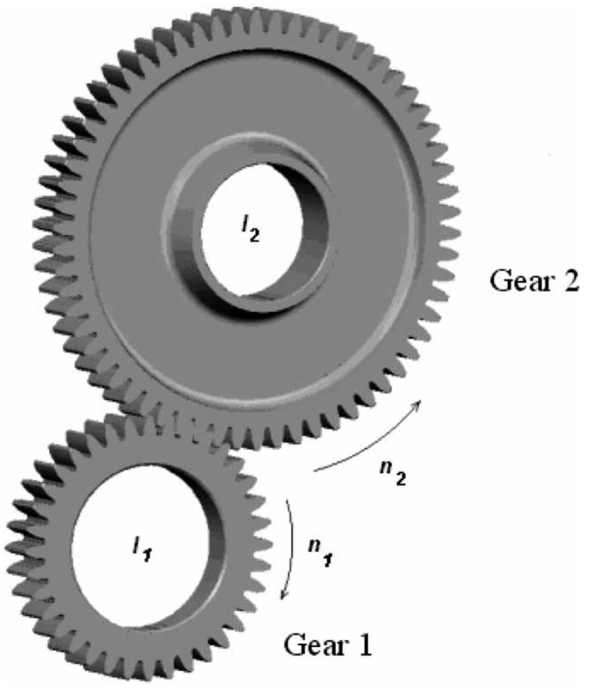

Fig. 7 Gear train inertia reduction (gear 1: crankshaft gear)

speed n_1 (e.g. a crankshaft gear) can be expressed as

$$I_{\text{red}} = I_2 \left(\frac{n_2}{n_1}\right)^2 \tag{3}$$

This reduction must be done for all the components activated by the gear train in relation to the crankshaft gear. Figure 7 illustrates an example.

3.2 Torsional stiffness

The torsional stiffness of all sections of the crankshaft model can be calculated considering finite element models, where a constant torque is applied at one side of the part and the twist angle is obtained considering that the model is clamped at the other extremity. The relation between the torque and the calculated twist angle is the torsional stiffness that is considered in the equivalent model (Fig. 8).

3.2.1 Rubber TVD

The dynamic stiffness of the rubber TVD shown in Fig. 9 was also determined based on a finite element model. For this calculation, the authors adopted a dynamic shear modulus of rubber in the range of 1.5 MPa $\leqslant G \leqslant$ 3.0 MPa, according to references [**13**] and [**14**]. Poisson's ratio is 0.49.

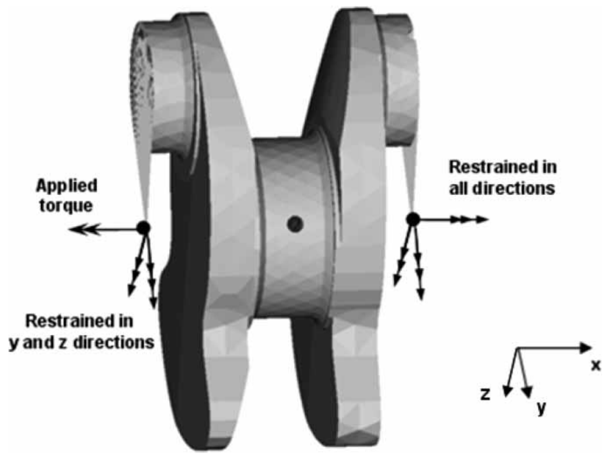

Fig. 8 Stiffness calculated by the finite element method (FEM)

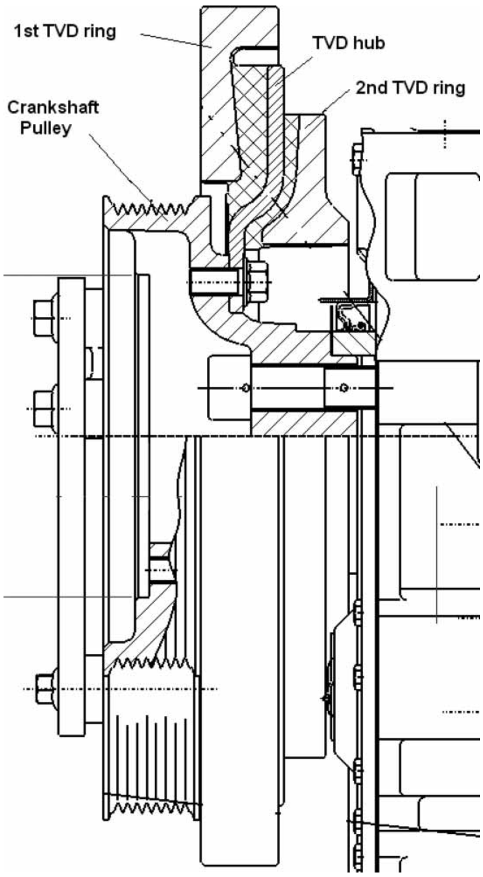

Fig. 9 Double mass rubber damper

3.2.2 Viscous TVD

The torsional stiffness of the viscous damper was determined according to the following methodology (see reference [**14**]) as a function of the silicone kinematic viscosity. The dynamic stiffness is

$$k_t = Gs\ S \tag{4}$$

Table 1 Determining factors of the viscous damper torsional stiffness according to reference [**14**]

Viscosity (m^2/s)	0.1	0.14	0.20
G_{01} (N/m^2)	21E–04	24E–04	105E–04
B_{01} (K)	3630	3821	3511
A_{01} (–)	2.28	2.37	2.15
a_{11} (K)	439	501	451

where

$$Gs = G_{01} e^{k_1} f^{k_2};\, k_1 = B_{01}/T;\, k_2 = a_{01} - a_{11}/T;\, f = n\, n_e$$

and where S is the clearance factor (m^3), obtained from the TVD manufacturer, T the absolute mean temperature (K) of the silicone film, n the order number, and n_e the engine speed (per second).

The constants are shown in Table 1. These parameters were obtained empirically in specific test beds, considering the variation of the silicone temperature and excitation frequency to determine the stiffness in each frequency step. Figure 10 gives a qualitative illustration of the variation of silicone torsional stiffness. Increasing the temperature causes the silicone stiffness to decrease. Figure 11 shows the main components of a viscous TVD.

3.3 Damping coefficients

The relative damping coefficients of the system, 'c_r', can be obtained from the loss angle property, as will be shown. The loss angle can be calculated by the following equation, considering that 'ω' is the engine's angular velocity

$$\chi = \tan\delta = \frac{c_r \omega}{k_t} \tag{5}$$

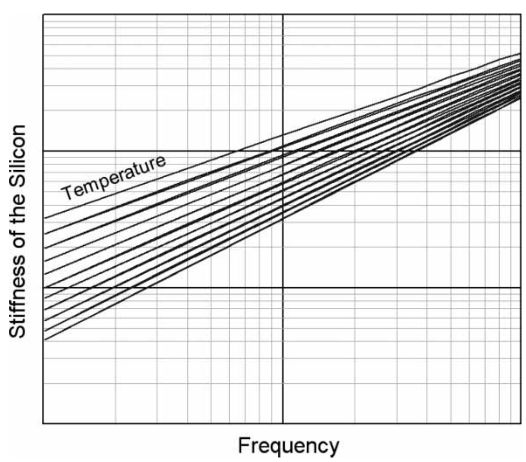

Fig. 10 Variation in viscous damper torsional stiffness (courtesy: Hasse&Wrede)

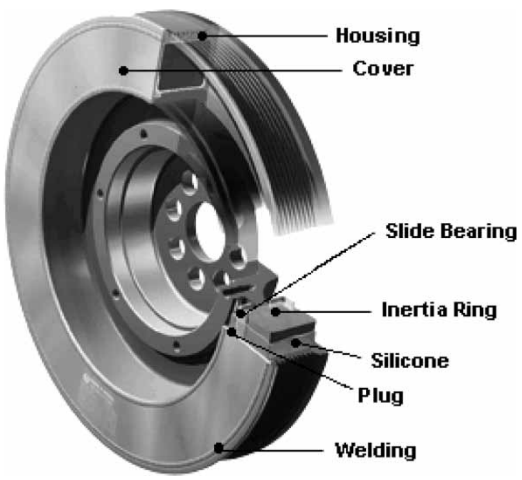

Fig. 11 Viscous damper details (courtesy: Hasse& Wrede)

Table 2 Four-stroke diesel engine average loss factors (TC, turbocharged engine)

Engine type	Loss factor (d)
In-line 4 cylinders (TC)	0.055
In-line 6 cylinders (TC)	0.035

At resonance, the loss factor property is defined as

$$d = \frac{c_r \cdot \omega_n}{k_t} \tag{6}$$

The average loss factor can be calculated according to the type of engine. Table 2 presents the common values for this property. See references [**14**] and [**15**] for other engine types. It is important to note that there is a different loss factor for each order of vibration, resulting in different damping coefficients.

Note that at a natural frequency, 'ω_n', the loss factor is equal to the loss angle, and considering the torsional stiffness, 'k_t', one can determine the relative damping coefficient.

The absolute damping coefficients considered in the position of the crank throw inertias are basically due to the contact between the piston rings and the block and oil films. It is advisable to determine these properties experimentally, running the engine without TVD and measuring the torsional vibration amplitudes with a dynamometer. The calculated vibration amplitudes should then be adjusted to the measured ones. In this specific case, the authors determined a value of 2.0 Nms/rad for this property.

3.3.1 Rubber TVD

To determine the rubber TVD's relative damping coefficient, a loss factor in the range of $0.15 \leqslant d \leqslant 0.25$ can be adopted, according to reference [**14**].

Table 3 Factors for determining the viscous damper damping coefficient, according to reference [**14**]

Viscosity (m^2/s)	0.10	0.14	0.20
G_{02} (N/m^2)	0.75	1.04	1.36
B_{02} (K)	2342	2373	2405
a_{02} (–)	1.49	1.51	1.55
a_{12} (K)	293	319	351

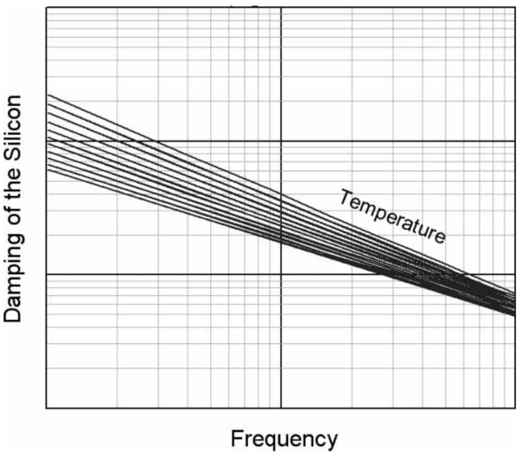

Fig. 12 Variation in silicone damping (courtesy: Hasse&Wrede)

3.3.2 Viscous TVD

The relative damping coefficient of the viscous damper is determined as follows

$$c_r = \frac{G\nu\, S}{\omega} \tag{7}$$

where $G\nu = G_{02} e^{k_3} f^{k_4}$; $k_3 = B_{02}/T$; $k_4 = a_{02} - a_{12}/T$; $\omega = 2\pi f$; $f = n\, n_e$. Here S is the clearance factor (m^3), obtained from the TVD manufacture, T the absolute mean temperature (K) of the silicone film, n the order number, and n_e the engine speed (s^{-1}). The constants are listed in Table 3.

These parameters are obtained as the same manner as explained in section 3.2 and the variation of the silicone damping as a function of the temperature and excitation frequency is shown, qualitatively, in Fig. 12. The higher the temperature the lower the silicone damping.

3.4 Excitation torque

The torque, which actuates the crank throws is calculated from the tangential force multiplied by the crank radius. Initially, the kinematics of the crank mechanism is determined for further dynamic loading computation. The methodology presented here is fully described in references [**16**] and [**17**]. Figure 13 indicates the main dimensions and the loads acting upon the crank mechanism.

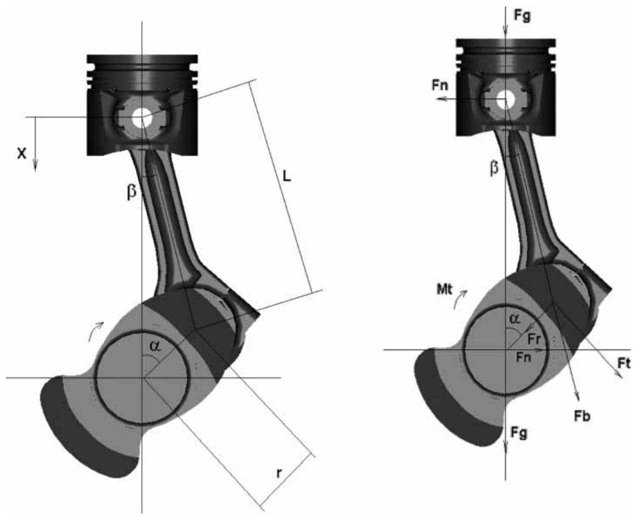

Fig. 13 Dimensions for kinematic and dynamic analyses

Only the tangential force, 'F_t', is computed for the TVA. The other loads, such as radial force, 'F_r', are important in the structural analysis of the crankshaft but those calculations are outside the scope of this work. The tangential force is calculated based on the gas load and the inertial forces of the system.

The gas load can be obtained by the equation

$$F_g = \frac{\pi d_p^2}{4} p \tag{8}$$

where d_p is the piston diameter and p the cylinder pressure: $p = p(\alpha)$.

The tangential gas load is computed as

$$F_{tg} = F_g \frac{\sin(\alpha + \beta)}{\cos\beta}; \ \sin\beta = \lambda \sin\alpha \tag{9}$$

The oscillating inertial force can be determined as follows, according to reference [**16**], considering terms up to the sixth order of the series. It is actually possible to disregard the terms with orders higher than the second one without compromising the accuracy of the results, taking into account the small values of the relation 'λ'.

$$F_{ia} = m_a r \omega^2 \left(\cos\alpha + \lambda \cos 2\alpha - \frac{\lambda^3}{4} \cos 4\alpha + \frac{9\lambda^5}{128} \cos 6\alpha \right); \qquad \lambda = \frac{r}{L} \tag{10}$$

Similarly, the tangential inertial force is

$$F_{ta} = F_{ia} \frac{\sin(\alpha + \beta)}{\cos\beta} \tag{11}$$

where m_a is the oscillating masses (complete piston 'm_p' plus con rod oscillating mass 'm_{ab}'), r the crank

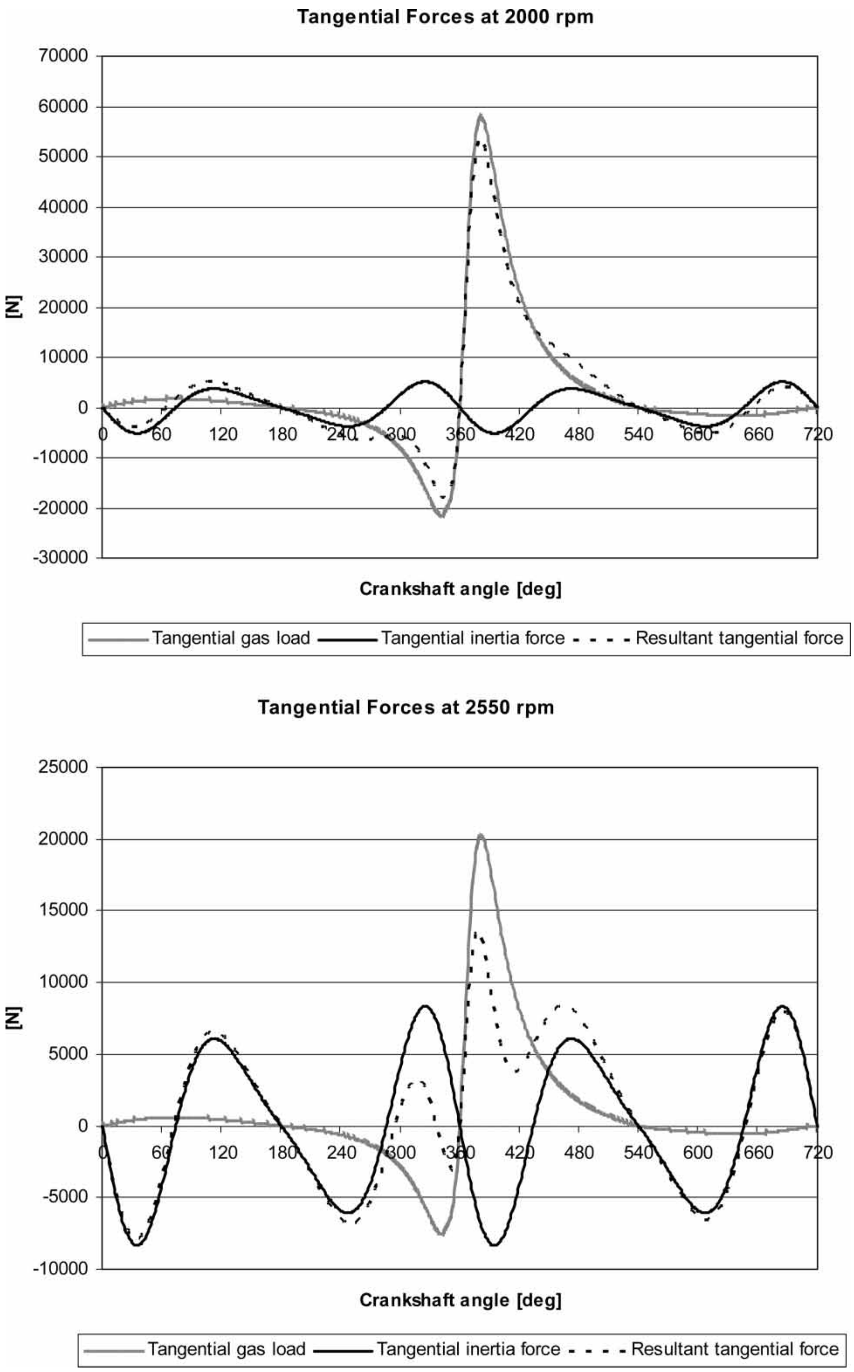

Fig. 14 Crankpin tangential forces at 2000 and 2550 r/min

radius, ω the angular velocity of the engine, L the con rod length, and α the crank angle.

Thus, the resulting tangential force is

$$\vec{F}_{\mathrm{t}} = \vec{F}_{\mathrm{tg}} + \vec{F}_{\mathrm{t}a} \tag{12}$$

To exemplify, Fig. 14 shows the variation of tangential forces as a function of the crank angle at engine speeds of 2000 and 2550 r/min. Note the influence of inertial forces on higher engine speeds.

Finally, the excitation torque can be determined simply by multiplying the resulting tangential force by the crankshaft radius

$$M_{\mathrm{t}} = F_{\mathrm{t}}\, r \tag{13}$$

3.5 Dynamic characteristics of the system

The differential equation of the system, representing the dynamic characteristics of mechanical vibrations, can be determined according to the procedures outlined below. More detailed information on this subject is given in references [**18**] to [**20**]

$$[\mathbf{M}]\ \{\ddot{\theta}(t)\} + [\mathbf{C}]\ \{\dot{\theta}(t)\} + [\mathbf{Kt}]\ \{\theta(t)\} = T(t) \tag{14}$$

The number of degrees of freedom of the system is equal to the number of inertias. Considering the equivalent system in Fig. 4, the matrices of equation (14) have the following expressions and, due to the lumped model considered here, they are band matrices

$$\text{Inertia matrix}: [\mathbf{M}] = \operatorname{diag}[I(j)]; \qquad j = 1(1)11$$

The absolute damping matrix, whose coefficients were determined as explained in the previous section, is diagonal and has non-zero values only in positions of the crank throws, i.e. from positions 5 to 10.

Thus, the total damping matrix can be obtained by the sum of previous matrices

$$[\mathbf{C}] = [\mathbf{Ca}] + [\mathbf{Cr}] \tag{16}$$

Due to the rigidity between inertias, the torsional stiffness matrix is similar to the relative damping matrix

$$[\mathbf{Kt}] = \begin{bmatrix}
kt(1) & 0 & -kt(1) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & kt(2) & -kt(2) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-kt(1) & -kt(2) & kt(3)+kt(2)+kt(1) & -kt(3) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -kt(3) & kt(4)+kt(3) & -kt(4) & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -kt(4) & kt(5)+kt(4) & -kt(5) & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -kt(5) & kt(6)+kt(5) & -kt(6) & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -kt(6) & kt(7)+kt(6) & -kt(7) & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -kt(7) & kt(8)+kt(7) & -kt(8) & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -kt(8) & kt(9)+kt(8) & -kt(9) & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -kt(9) & kt(10)+kt(9) & -kt(10) \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -kt(10) & kt(10)
\end{bmatrix}$$

The oscillating masses are replaced by equivalent inertias, which must have the same kinetic energy as the piston motion. An average inertia is used for the calculations, during one revolution of the crankshaft. Equation (15) quantifies this inertia, which is introduced only in the crank throw matrix positions

$$I_{\text{alt}} = m_a r^2 \left(\frac{1}{2} + \frac{\lambda^2}{8}\right) \tag{15}$$

The relative damping matrix depicts the coupling between the rubber TVD in the first rows and columns of the matrix and the crankshaft, represented by the terms from indexes 4 to 10. The double mass rubber TVD configuration modifies the first terms of the matrix, as follows

As mentioned before, the excitation torque actuating the crankshaft varies according to the crank angle, engine speed, and engine load

$$\{T(t)\} = \begin{matrix} \{0 & 0 & 0 & 0 & \ldots \\ Mt^1(t) & Mt^2(t) & Mt^3(t) & Mt^4(t) & \ldots \\ Mt^5(t) & Mt^6(t) & 0\}^{\mathrm{T}} & & \end{matrix}$$

The torque, $M_{\mathrm{t}}^{q}(t)$, which actuates each crank throw is a periodic excitation function displaced in time by an amount that depends on the engine ignition sequence. The solution for this kind of system is found through a finite Fourier series, see reference [**21**]. In this study, the authors considered 24 terms for the

$$[\mathbf{Cr}] = \begin{bmatrix}
cr(1) & 0 & -cr(1) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & cr(2) & -cr(2) & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-cr(1) & -cr(2) & cr(3)+cr(2)+cr(1) & -cr(3) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & -cr(3) & cr(4)+cr(3) & -cr(4) & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -cr(4) & cr(5)+cr(4) & -cr(5) & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -cr(5) & cr(6)+cr(5) & -cr(6) & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & -cr(6) & cr(7)+cr(6) & -cr(7) & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & -cr(7) & cr(8)+cr(7) & -cr(8) & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -cr(8) & cr(9)+cr(8) & -cr(9) & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -cr(9) & cr(10)+cr(9) & -cr(10) \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -cr(10) & cr(10)
\end{bmatrix}$$

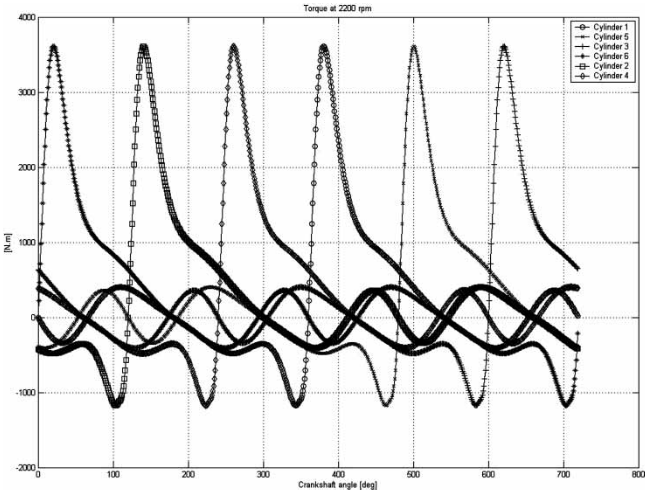

Fig. 15 Instantaneous torque on the crankshaft at 2200 r/min

series expansion.

$$M_{\mathrm{t}}^{q}(t) = \frac{A_0^q}{2} + \sum_{n=1}^{24}\left[C_n^q\,\mathrm{e}^{i\,n\,\omega\,t} + \overline{C}_n^q\,\mathrm{e}^{-i\,n\,\omega\,t}\right]; \quad q = 1(1)6 \tag{17}$$

where $C_n^q = (1/2)(A_n^q - iB_n^q)$ and $\overline{C}_n^q = (1/2)(A_n^q + iB_n^q)$

The actuating moments in each crank throw are illustrated graphically in Fig. 15.

3.6 State equation solution

The dynamic behaviour of the crankshaft can be expressed through the system's first-order differential state equation

$$\dot{x}(t) = \mathrm{A}\,x(t) + b(t); \qquad \dot{x}(t) = \begin{Bmatrix}\theta(t)\\ \dot{\theta}(t)\end{Bmatrix} \tag{18}$$

where

$$\mathbf{A} = \begin{bmatrix} 0 & \mathrm{I} \\ -\mathbf{M}^{-1}\,\mathbf{Kt} & -\mathbf{M}^{-1}\,\mathbf{C}\end{bmatrix} \text{ and } \boldsymbol{b}(t) = \begin{Bmatrix} 0 \\ \mathrm{M}^{-1}\,T(t)\end{Bmatrix}$$

Using the representation of $M_t^q(t)$ in the frequency domain, equation (17), the excitation vector $\{b(t)\}$ can be calculated according to the equation

$$\boldsymbol{b}(t) = \frac{b_o}{2} + \sum_{n=1}^{24}\left[b_n\cdot\mathrm{e}^{i\cdot n\omega\cdot t} + \overline{b_n}\cdot\mathrm{e}^{-i\cdot n\omega\cdot t}\right] \tag{19}$$

where

$$b_o = \left\{\{0\}_{n\times 1}\vdots\ 0\ \ 0\ \ 0\ \ 0\ \ \frac{A_0^1}{I(5)}\ \ \frac{A_0^2}{I(6)}\ \ \frac{A_0^3}{I(7)}\ \ \frac{A_0^4}{I(8)}\ \ \frac{A_0^5}{I(9)}\ \ \frac{A_0^6}{I(10)}\ \ 0\right\}^{\mathrm{T}}$$

$$b_n = \left\{\{0\}_{n\times 1}\vdots\ 0\ \ 0\ \ 0\ \ 0\ \ \frac{C_n^1}{I(5)}\ \ \frac{C_n^2}{I(6)}\ \ \frac{C_n^3}{I(7)}\ \ \frac{C_n^4}{I(8)}\ \ \frac{C_n^5}{I(9)}\ \ \frac{C_n^6}{I(10)}\ \ 0\right\}^{\mathrm{T}}$$

$$\overline{b_n} = \left\{\{0\}_{n\times 1}\vdots\ 0\ \ 0\ \ 0\ \ 0\ \ \frac{\overline{C_n}^1}{I(5)}\ \ \frac{\overline{C_n}^2}{I(6)}\ \ \frac{\overline{C_n}^3}{I(7)}\ \ \frac{\overline{C_n}^4}{I(8)}\ \ \frac{\overline{C_n}^5}{I(9)}\ \ \frac{\overline{C_n}^6}{I(10)}\ \ 0\right\}^{\mathrm{T}}$$

3.7 Calculation of the system's steady-state response

The response of a periodic excited vibratory linear system, represented by its state equation, can be obtained via the fundamental matrix, or transition state matrix and the convolution integral

$$x(t) = \mathbf{\Phi}(t)\,x(0) + \frac{1}{2}\int_0^t \mathbf{\Phi}(t-\tau)\,b_0\,\mathrm{d}\tau + \sum_{n=1}^{24}\int_0^t \mathbf{\Phi}(t-\tau)\,(b_n\,\mathrm{e}^{in\omega\tau} + \overline{b}_n\,\mathrm{e}^{-in\omega\tau})\mathrm{d}\tau \tag{20}$$

where $\Phi(t) = \mathrm{e}^{A\cdot t}$

Disregarding the transitory and constant Fourier term and solving the summation of the harmonic terms, the steady-state response can be obtained as follows

$$x_n(t) = \theta_n(t) = g_n\,\mathrm{e}^{in\omega t} + \overline{g}_n\,\mathrm{e}^{-in\omega t} \tag{21}$$

where the frequency response vectors are $g_n = \mathrm{F}_n\cdot b_n$ and $\overline{g_n} = \overline{\mathrm{F}_n}\cdot\overline{b_n}$ and the frequency matrices are $\mathbf{F}_n = (i\ n\ \omega\,\mathrm{I} - \mathrm{A})^{-1}$ and $\overline{\mathrm{F}_n} = (-i\cdot n\cdot\omega\cdot\mathrm{I} - \mathrm{A})^{-1}$

Therefore, the global vibration amplitude can be computed by the following equation

$$\theta_j = \sum_{n=1}^{24}\Theta_{n_j}\ \cos(n\ \omega\ t - \phi_{n_j}) \tag{22}$$

where $\Theta_{n_j} = 2\sqrt{[\mathrm{Re}(g_{n_j})]^2 + [\mathrm{Im}(g_{n_j})]^2} = 2\left|g_{n_j}\right|$; $\phi_{n_j} = \mathrm{a}\tan(-\mathrm{Im}(g_{n_j})/\mathrm{Re}(g_{n_j}))$; $n = 1(1)24$; $j = 1(1)11$.

Knowing the torsional vibration amplitude of two consecutive inertias, the actuating dynamic torque can be calculated according to the following equation

$$T_{j-1} = \left|\theta_j - \theta_{j-1}\right|\ kt_{j-1}; \quad j = 1(1)11 \tag{23}$$

It is important to note that the constant Fourier term must be added to the calculated torsional vibration torque, taking into account the number of cylinders ahead of the considered inertia. For example, the constant Fourier term must be added six times to the calculated torque between the flywheel and the sixth cylinder.

From the TVA, one can calculate the dissipated energy at the TVD. The damper thermal load is given by

$$Q_j = \int_0^t cr_j \left(\dot{\theta}_j - \dot{\theta}_3\right)^2 \mathrm{d}t; \quad j = 1, 2 (\text{double mass rubber TVD}) \tag{24}$$

$$Q_1 = \int_0^t cr_1 \left(\dot{\theta}_1 - \dot{\theta}_2\right)^2 \mathrm{d}t (\text{single mass viscous TVD}) \tag{25}$$

The permissible dissipated power for a rubber damper can be calculated according to the following methodology.

The mean convection coefficient at the damper's external faces can be computed according to reference [**14**], as follows

$$h_c = 7.56 \left(\frac{\pi D n_\mathrm{e}}{60}\right)^{0.8} (\mathrm{W/m^2K}) \tag{26}$$

where D is the diameter for convection coefficient evaluation (m) and n_e the engine speed (r/min).

Applying this thermal load to a finite element model and considering the thermal conductivity of 0.26 W/m K for the rubber, one can determine the maximum power that the damper can dissipate, taking

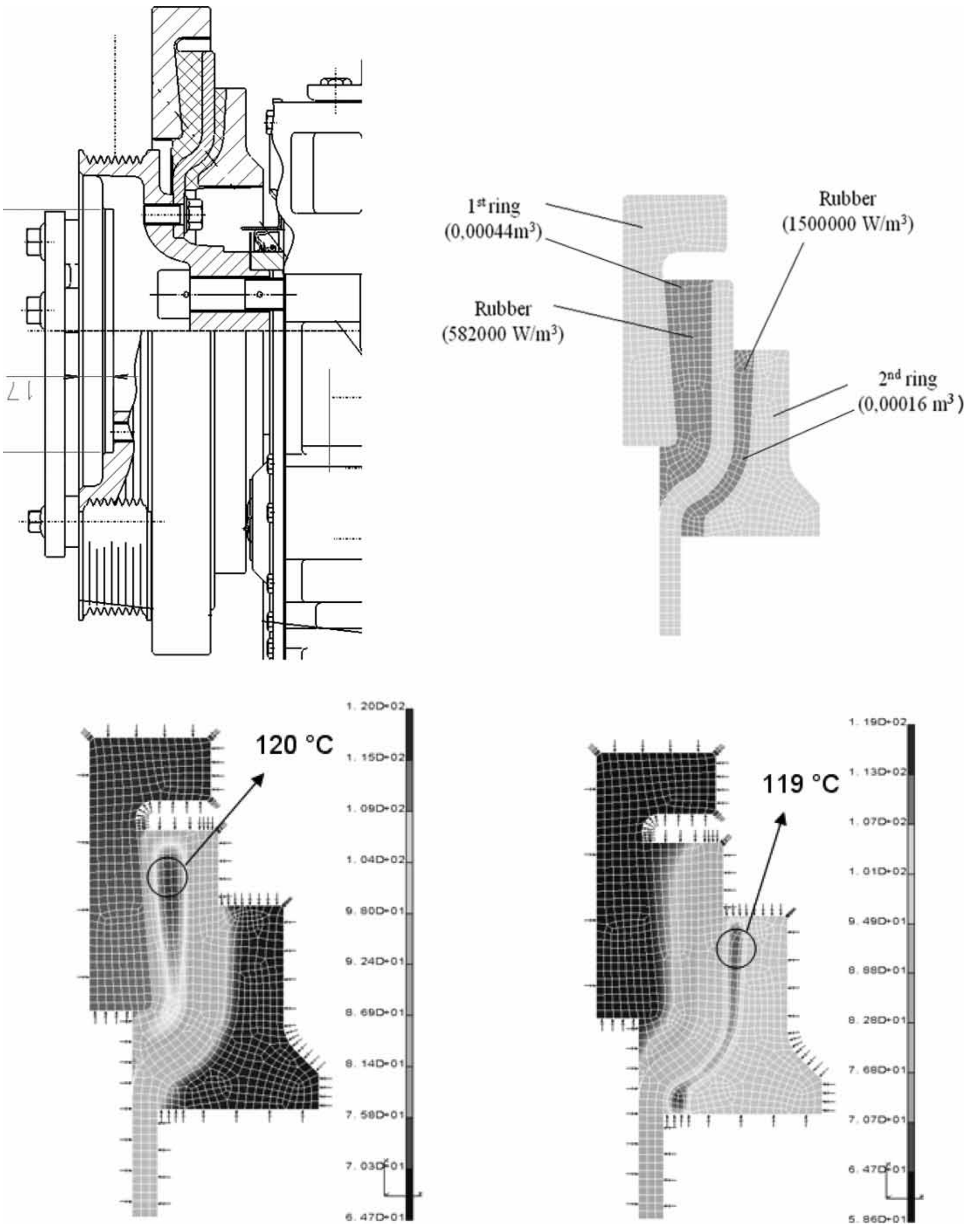

Fig. 16 FEM with axisymmetric solid elements, and results of thermal analyses

into account that 120 °C is the maximum operational temperature for nitrile butadiene rubber (NBR).

The thermal analysis also considers that the heat generated in the rubber damper rings is 582 000 W/m^3 in the 1st ring and 1 500 000 W/m^3 in the 2nd ring. Our analyses indicate that the permissible damper load is approximately 250 W at each damper ring. Figure 16 shows the boundary conditions and results of the heat transfer analysis.

In the case of viscous TVD, the permissible dissipated power in 'W' can be calculated according to the Iwamoto equation [**9**]

$$\dot{Q}_{\text{perm}} = f\, 105\, a_m\, A_d^{1.3} \left(\frac{2\,\pi\, n_{\text{e}}}{60}\right)^{0.8} (t_o - t_{\text{amb}}) \tag{27}$$

where $f = 1.23$ to 1.33 for dampers with cooling fins, otherwise $f = 1.0$; A_d is the reference area of the TVD ring (m^2), obtained from the TVD manufacturer; a_m the damper size factor: 0.0201–0.0303, see reference [**14**]; n_{e} the engine speed (r/min); t_o the temperature at TVD surface (°C); t_{amb} the ambient temperature (°C).

For a rubber TVD, one can also calculate the actuating shear stress and maximum deformation of the rubber. The maximum shear stress, which should not exceed 0.3 to 0.4 MPa, can be calculated from the relation between the torque at the damper ring and hub, taking into account the rubber section modulus under shear

$$\tau_j = \frac{\left|\theta_j - \theta_3\right|\, k_{t_j}}{W_{t_j}};\ j = 1, 2 (\text{for a double mass TVD}) \tag{28}$$

The maximum deformation of the rubber, which should not exceed 15–20 per cent, can be calculated by the following equation, considering that for small angles, $\tan(\Delta\theta) \cong \Delta\theta$

$$\varepsilon_j = \frac{\tau_j \max W_{tj}}{k_{tj}}\, \frac{R_j}{e_j}\, 100\%; \quad j = 1, 2 \tag{29}$$

where W_t is the rubber section modulus under shear, k_{t} the rubber torsional stiffness, θ the torsional vibration amplitude, R the maximum radius of the rubber at TVD, and e the rubber thickness.

These permissible parameters are stipulated by TVD manufacturers and their reliability is verified through dynamometer and vehicle tests.

It is advisable to calculate the stress and strain of complex rubber geometries considering non-linear FE models. Thus, the stress concentration factor of any rubber geometry can be evaluated and this factor introduced into equation (28). Figure 17 exemplifies the principal stresses in the rubber for a given relative angular displacement of 1°.

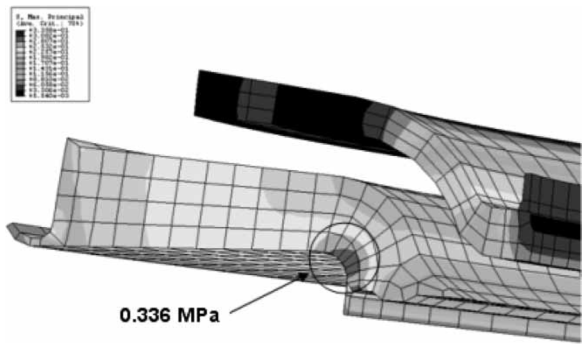

Fig. 17 Principal stress in TVD rubber

4 RESULTS AND DISCUSSION

This section presents the input data of the analysed systems and the results of the torsional vibration calculations made with the MATLAB® software.

(1) Dynamic characteristics of a system without TVD (based on Fig. 4):
  (a) Inertias (kg m^2)
    $I(3) = 0.0170$ (crankshaft pulley)
    $I(4) = 0.0090$ (gear train)
    $I(5) = 0.0467$ (1st crank throw and oscillating masses)
    $I(6) = 0.0327$ (2nd crank throw and oscillating masses)
    $I(7) = 0.0467$ (3rd crank throw and oscillating masses)
    $I(8) = 0.0467$ (4th crank throw and oscillating masses)
    $I(9) = 0.0327$ (5th crank throw and oscillating masses)
    $I(10) = 0.0487$ (6th crank throw and oscillating masses)
    $I(11) = 2.0750$ (flywheel and dynamometer coupling)
  (b) Torsional stiffness (Nm/rad)
    $k_{\text{t}}(3) = 1\,106\,000$
    $k_{\text{t}}(4) = 1\,631\,000$
    $k_{\text{t}}(5) = 1\,253\,000$
    $k_{\text{t}}(6) = 1\,253\,000$
    $k_{\text{t}}(7) = 1\,678\,000$
    $k_{\text{t}}(8) = 1\,253\,000$
    $k_{\text{t}}(9) = 1\,253\,000$
    $k_{\text{t}}(10) = 1\,976\,000$
  (c) Absolute damping (Nms/rad)
    $c_a(j) = 2.0; j = 5(1)10$
  (d) Relative damping
    Engine mean loss factor: $d = 0.035$
  (e) General data considered in the analyses:
    Constant gear train torque: 86 Nm
    Permissible front-end torque: 2012 Nm
    Permissible rear-end torque: 5413 Nm

(2) Dynamic characteristics of the system considering the viscous TVD (based on Fig. 3):
  (a) Inertia (kgm^2)
    $I(1) = 0.1520$ (TVD ring)
    $I(2) = 0.0970$ (TVD hub and crankshaft pulley)
  (b) Torsional stiffness (Nm/rad)
    $k_t(1)$ = calculated according to section 3.2
  (c) General TVD data:
    Kinematic viscosity of the silicone: $\nu = 0.2\,\mathrm{m^2/s}$
    Clearance factor: $S = 5.0\,\mathrm{m^3}$
    Damper size factor: $a_m = 0.025$
    Reference area of the TVD ring: $A_d = 0.1396\,\mathrm{m^2}$
    Silicone film maximum temperature: $t_{SIL} = 115\,°C$
    TVD maximum temperature: $t_o = 100\,°C$
    Ambient temperature: $t_{amb} = 51\,°C$
  (d) Note: considering a TVD with cooling fins for better heat dissipation:
    $\dot{Q}_{perm}$ is 23 per cent higher.
(3) Dynamic characteristics of the system with the double mass rubber TVD (based on Fig. 4):
  (a) Inertias (kgm^2)
    $I(1) = 0.1230$ (TVD 1st ring)
    $I(2) = 0.0273$ (TVD 2nd ring)
    $I(3) = 0.0440$ (TVD hub and crankshaft pulley)
  (b) Torsional stiffness (Nm/rad)
    $k_t(1) = 70\,000$ (TVD 1st ring)
    $k_t(2) = 88\,000$ (TVD 2nd ring)
  (c) Relative damping
    Rubber loss factor: $d = 0.15$
  (d) General TVD data:
    Rubber volume (1st ring): $0.00044\,\mathrm{m^3}$
    Rubber volume (2nd ring): $0.00016\,\mathrm{m^3}$
    Section modulus under shear (1st ring): $3.809 \cdot 10^{-3}\,\mathrm{m^3}$
    Section modulus under shear (2nd ring): $2.727 \cdot 10^{-3}\,\mathrm{m^3}$

All the analyses considered the measured combustion pressure curves to determine the excitation torque in the system. Figures 18 and 19 illustrate the variation of the cylinder pressure versus the crank angle and the peak cylinder pressure versus the engine speed, respectively. Qualitatively, the combustion pressure curves are similar at all engine speeds.

Figure 20 presents the results of the theoretical torsional vibration calculations considering no TVD assembled to the crankshaft. A comparison of the calculated and measured amplitudes in Fig. 21 enables one to adjust the actual absolute damping coefficients of the engine. All the figures show only the main orders of vibration for an in-line six cylinder engine, but the calculations were performed considering all 24 orders.

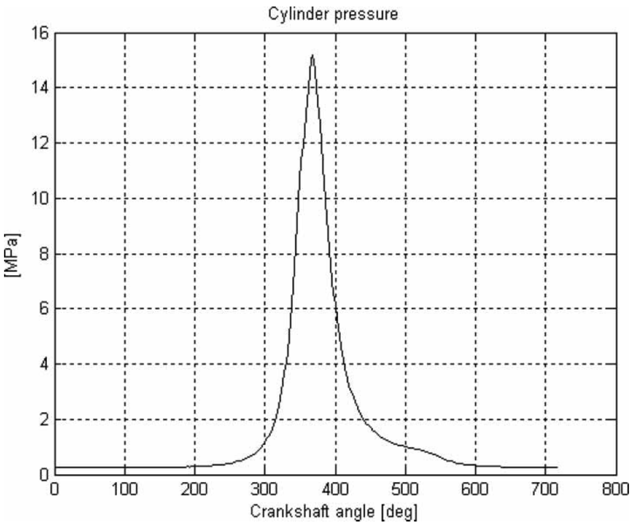

Fig. 18 Cylinder pressure curve at 2200 r/min

The nomenclature (6/I), as an example, represents the maximum vibration amplitude of the 6th order in the resonance of the 1st mode.

The figures below present results based on the data previously shown, considering the viscous damper to reduce the amplitudes of torsional vibration (Fig. 22).

The vibrations decreased considerably with this type of absorber – thanks to its higher damping capacity. Note that is difficult to identify the resonance frequencies in these figures (Fig. 23).

Lastly, the torsional vibrations analyses are presented considering the double mass rubber damper. With this type of TVD, the first two resonance frequencies excited by the main orders are clearly visible (Fig. 24).

An analysis of Fig. 25 and a comparison with the previous figure reveals that the measured and calculated

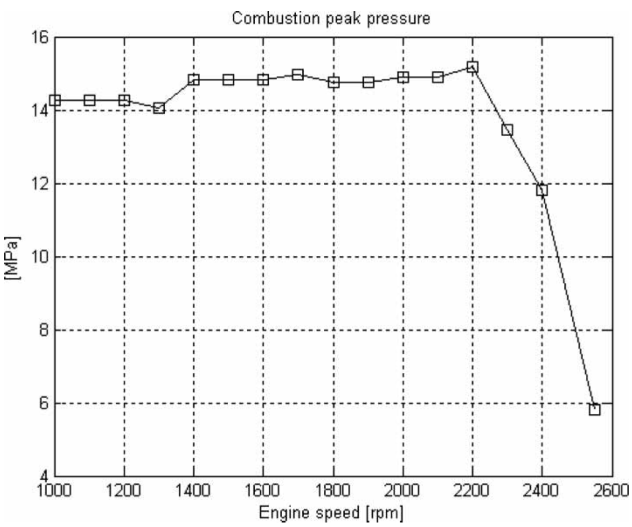

Fig. 19 Peak cylinder pressure at several engine speeds

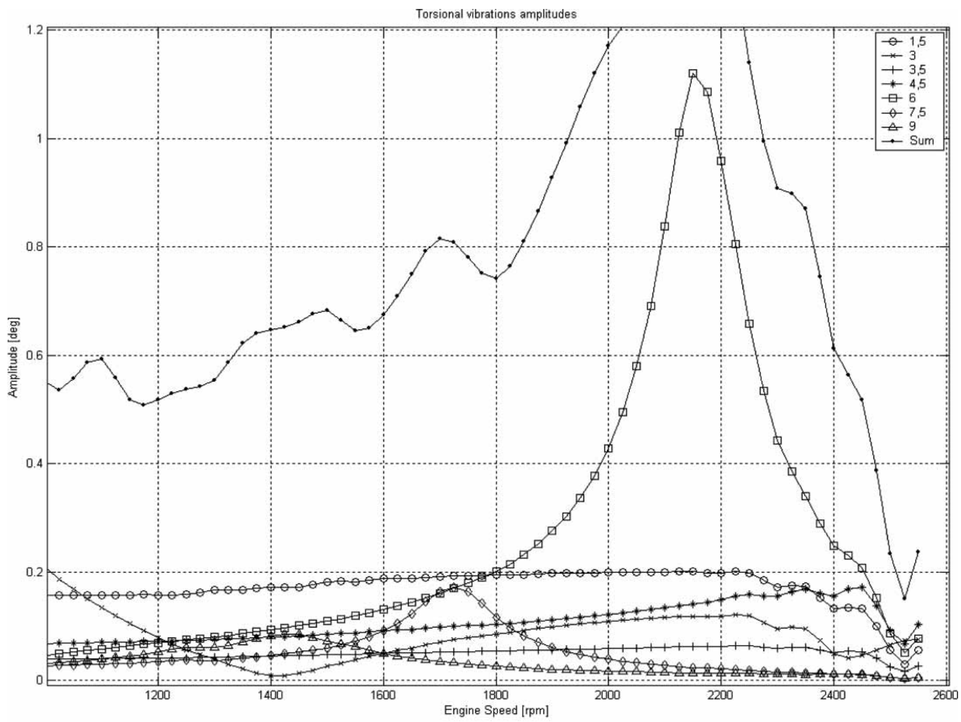

Fig. 20 Calculated torsional vibration amplitudes at crankshaft pulley without TVD

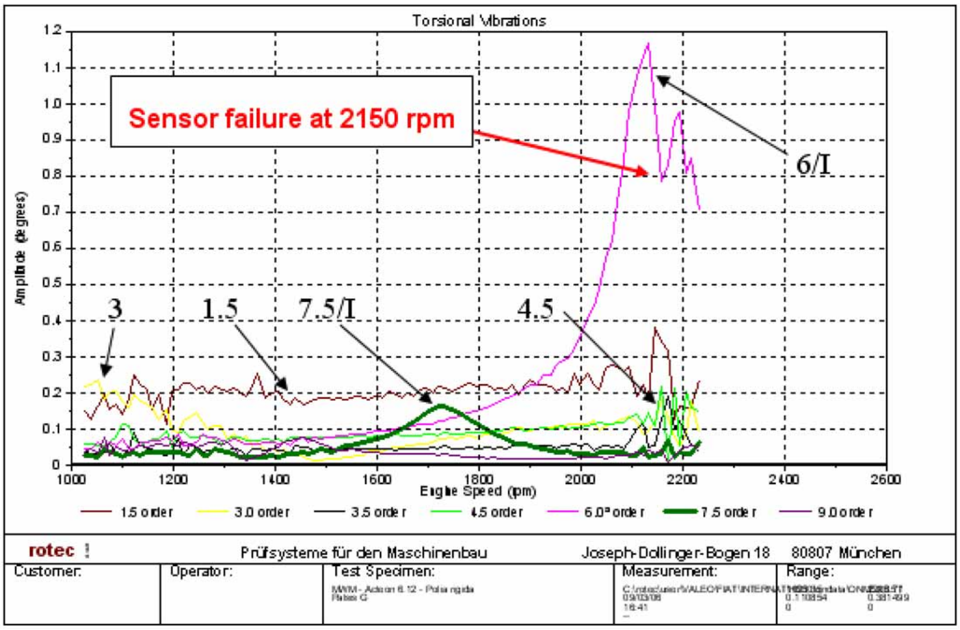

Fig. 21 Measured torsional vibration amplitudes in the crankshaft pulley without TVD (courtesy: Onça S.A.)

amplitudes have almost the same values and shapes. This comparison can be done for the other systems (w/o TVD and viscous TVD), indicating that the assumptions adopted for the methodology developed here are valid.

Figures 26 and 27 show the power generated in the rubber for the first and second damper rings. The influence of two modes of vibration, excited by 3rd and 6th orders at almost the same engine speed, produces a very high TVD load.

The shear stress and maximum rubber deformation calculated for both damper rings are shown in Figs 28 and 29. Comparing the maximum values to the permissible ones, one can conclude that this type of absorber is not suitable for the analysed engine.

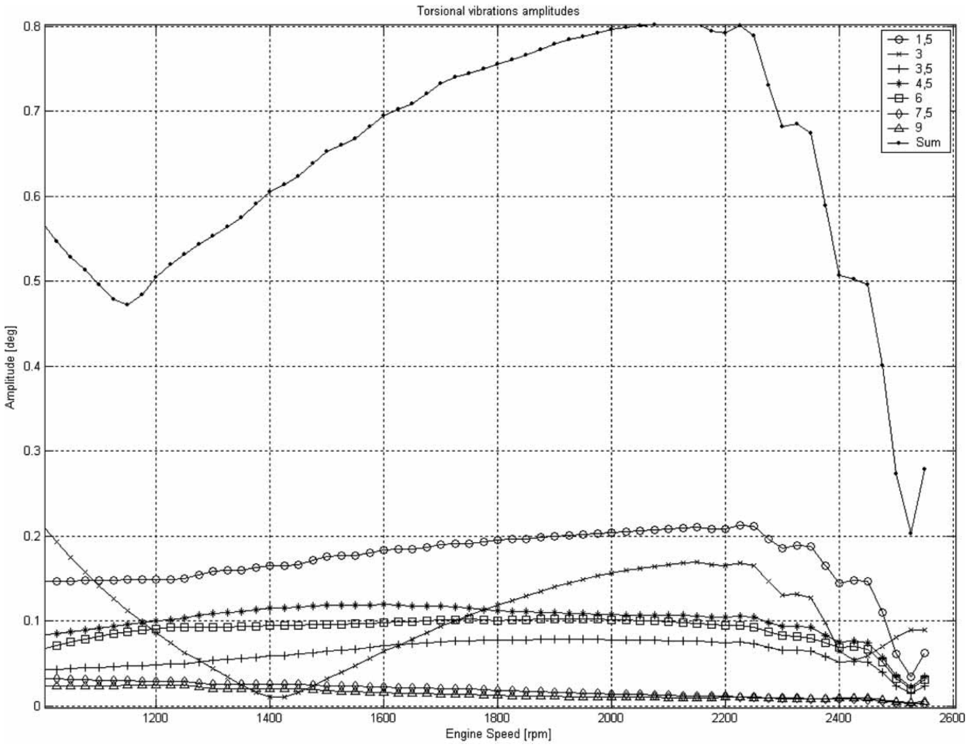

Fig. 22 Calculated amplitudes of torsional vibration in the crankshaft pulley with viscous TVD

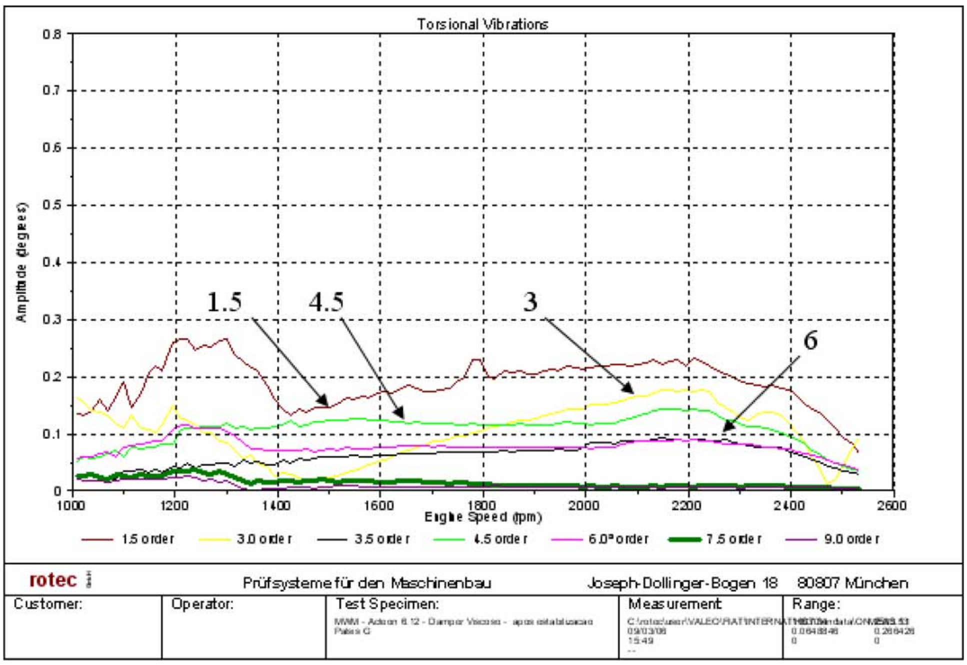

Fig. 23 Measured amplitudes of torsional vibration in the crankshaft pulley with viscous TVD (courtesy: Onça S.A.)

According to noise level and structural integrity design criteria, the maximum recommended vibration amplitudes, per order, in the crankshaft front-end should be in the range of 0.20° to 0.25° for in-line six cylinders engines. Considering the results presented here for a double mass rubber damper, one can see that the 3rd order/1st mode (3/I) and 6th order/2nd mode (6/II) have amplitudes exceeding 0.30°. The

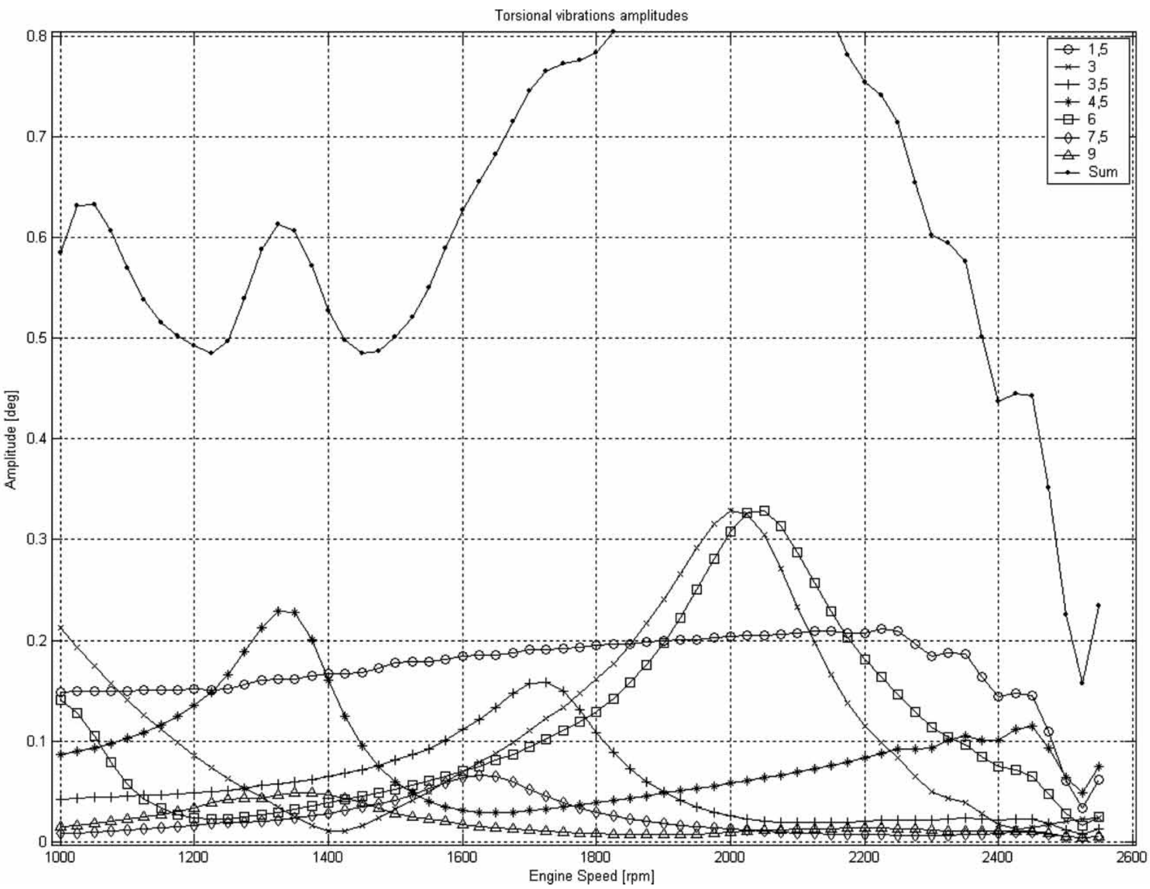

Fig. 24 Calculated amplitudes of torsional vibration in the crankshaft pulley with rubber TVD

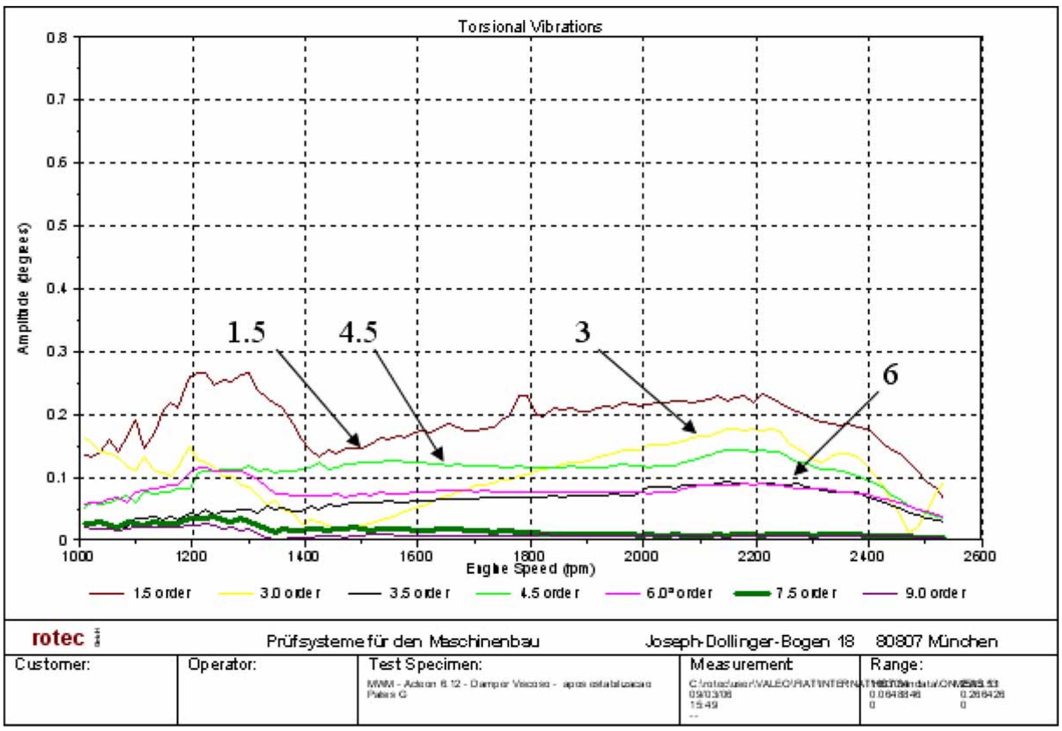

Fig. 25 Measured amplitudes of torsional vibration in the crankshaft pulley with rubber TVD (courtesy: Onça S.A.)

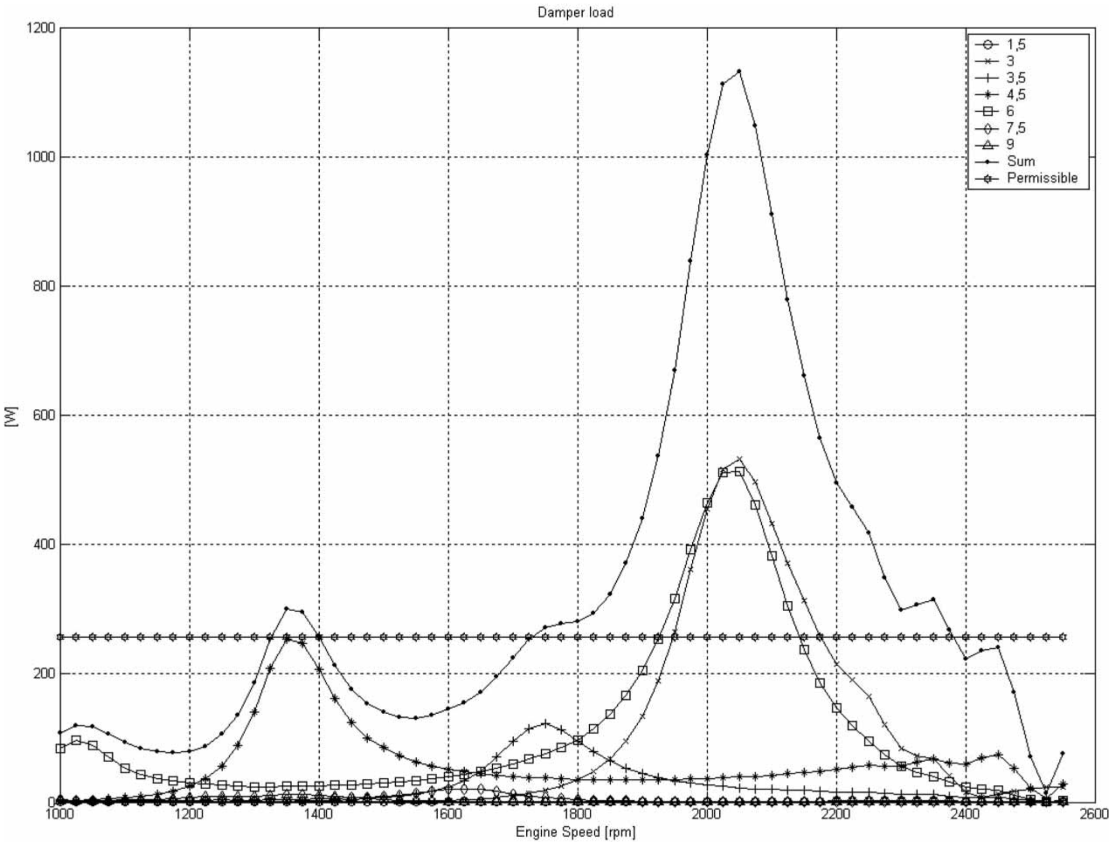

Fig. 26 Rubber damper load (1st ring)

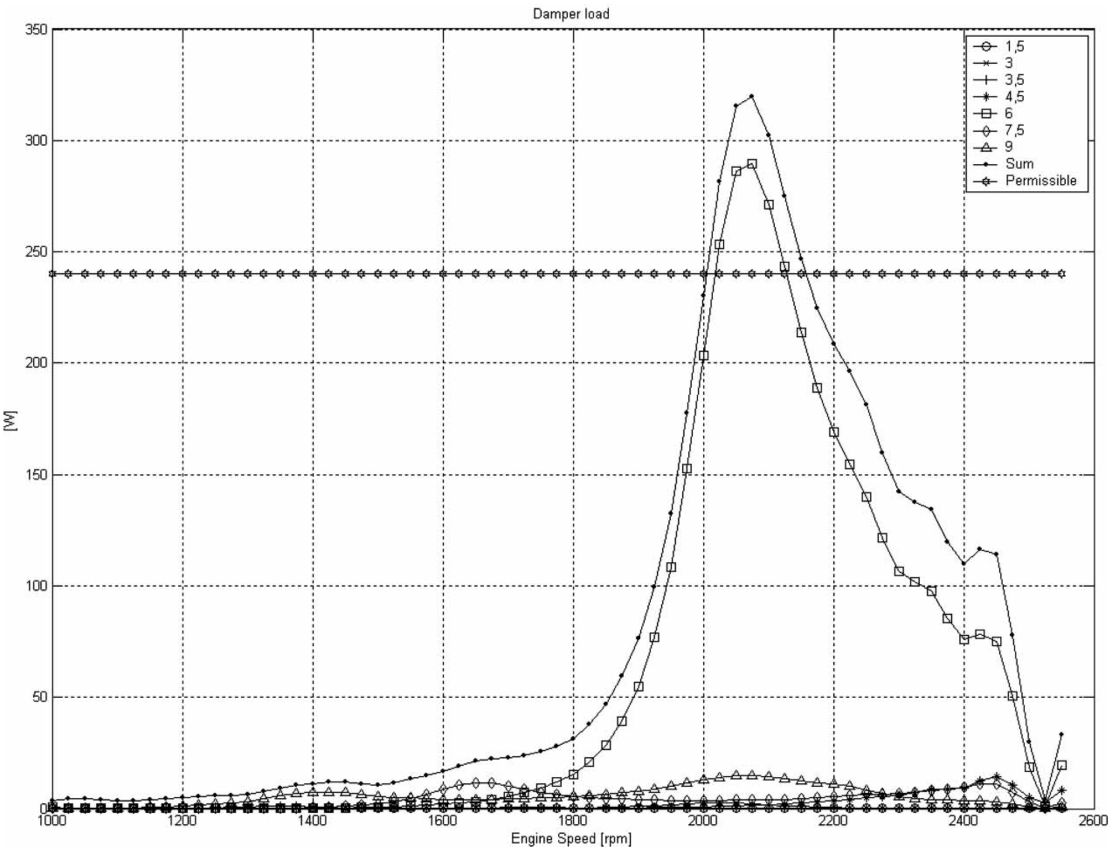

Fig. 27 Rubber damper load (2nd ring)

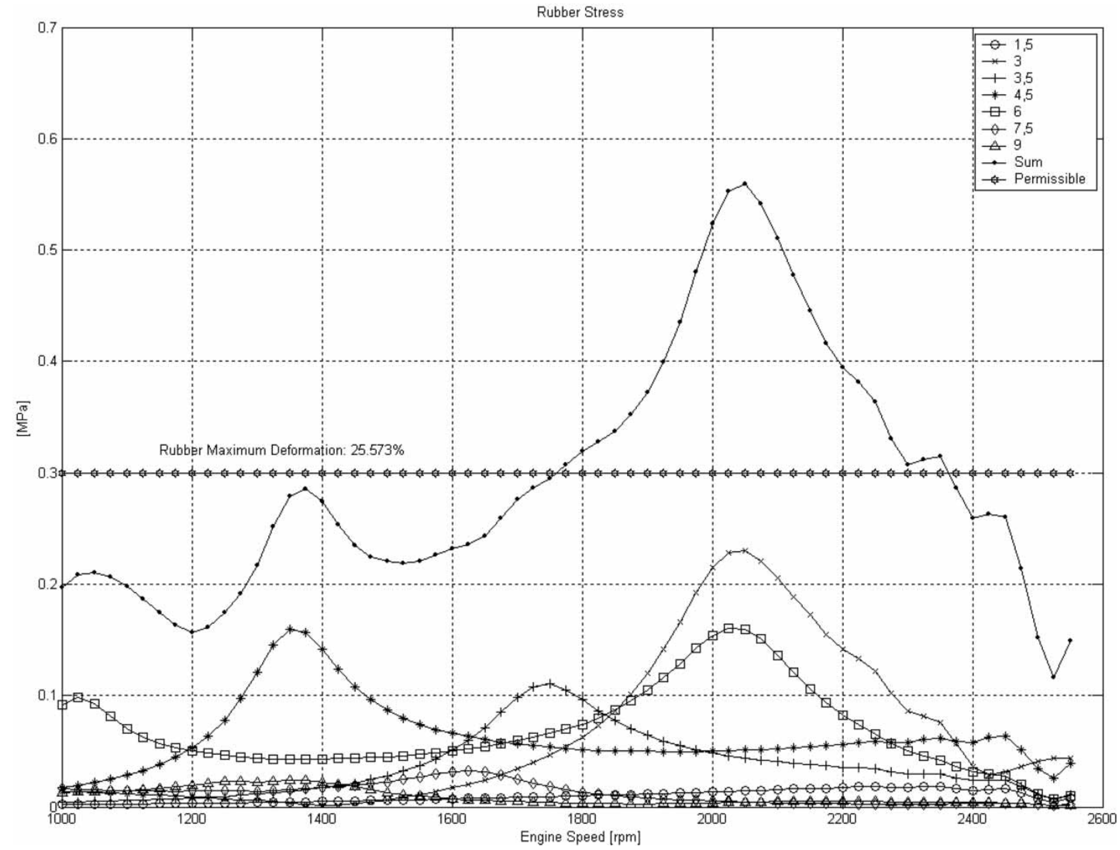

Fig. 28 Rubber shear stress (1st ring)

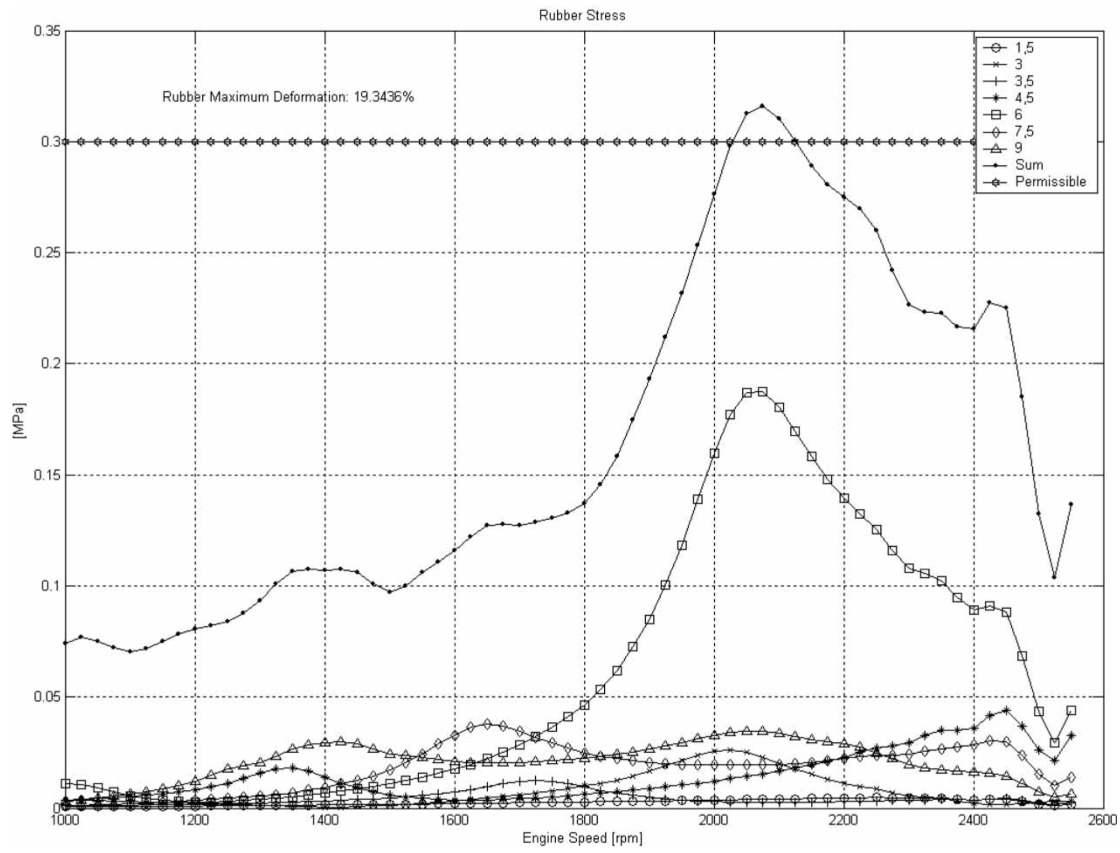

Fig. 29 Rubber shear stress (2nd ring)

Fig. 30 Structural failure of 1st TVD ring due to overload

maximum dissipated power is close to 1100 W in the 1st TVD ring and 325 W in the second. For this type of component, the permissible continuous damper load is about 250 W. The shear stress and maximum rubber deformation are above the recommended limits.

Therefore, the rubber TVD is not recommended for the engine in question. Only the viscous damper is suitable for the aforementioned application in terms of design criteria. Figure 30 shows structural failure of the rubber damper which occurred in a dynamometer test at a critical engine speed, i.e. close to 2100 r/min.

Appendix 2 shows additional results of the TVA.

5 CONCLUSIONS

An analysis of the results obtained and comparison with the measured results leads to the conclusion that the proposed methodology for TVA presents similar results. The hypotheses adopted for determining the equivalent model are therefore valid.

This technique allows for the determination of new design parameters, which could be optimized with shorter development times and fewer tested parts, thus offering in an attractive technical and commercial proposals.

The calculation methodology presented here can be applied to several types of ICEs from spark ignition to diesel engines, in-line or 'V' types, and 2- or 4-stroke engines, taking into account the correct ignition timing and sequence. However, for large displacement engines, e.g. marine ICE, other effects such as crankshaft axial vibrations and the influence of large oscillating parts cannot be disregarded in the calculations.

The software for TVA was specially developed in MATLAB® to be applied in new designs of TVD, considering the hardware-in-the-loop (HIL) technique, which can considerably reduce the cost involved in component durability validations, precluding the need for vehicle and/or dynamometer tests. The absorber is considered the hardware of the HIL model and this technique is currently under investigation by the authors.

The inclusion of axial and flexural vibrations in the proposed model can be also seen as a next step in this study, considering that, in some particular cases, axial vibrations in the system cannot be neglected.

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APPENDIX 1

Notation

$\mathbf{A}$	state matrix (–)
A_d	reference area of TVD ring (m^2)
$\mathbf{C}$	total damping matrix (Nm s/rad)
$\mathbf{Ca}$	absolute damping matrix (Nm s/rad)
$\mathbf{Cr}$	relative damping matrix (Nm s/rad)
d	loss factor (–)
d_p	piston diameter (m)
F_b	connecting rod force (N)
F_g	gas load (N)
F_{ia}	oscillating inertial force (N)
F_t	resulting tangential force (N)
F_{ta}	tangential oscillating force (N)
F_{tp}	tangential gas load (N)
G	dynamic shear modulus (MPa)
I	moment of inertia (kgm^2)
I_{alt}	moment of inertia of oscillating masses (kgm^2)
I_{red}	reduced moment of inertia (kgm^2)
j	degree of freedom (–)
$\mathbf{Kt}$	torsional stiffness matrix (Nm/rad)
L	connecting rod length (m)
L_1	distance from connecting rod centre of gravity to smaller ring (m)
L_2	distance from connecting rod centre of gravity to larger ring (m)
m_a	oscillating masses (kg)
m_{ab}	con rod oscillating mass (kg)
m_b	con rod total mass (kg)
m_{rb}	con rod rotating mass (kg)
$\mathbf{M}$	inertia matrix (kg m^2)
M_t	torque (Nm)
n	order number (–)
n_e	engine speed (r/min)
p	cylinder pressure (bar)
q	cylinder number (–)
Q	damper dissipated energy (J)
$\dot{Q}$	damper dissipated power (W)
r	crankshaft radius (m)
s	piston stroke (m)
S	viscous damper clearance factor (m^3)
Wt	rubber section modulus under shear (m^3)
α	crankshaft angle (degrees)
β	connecting rod angle (degrees)
δ	loss angle (rad)
ε	rubber deformation (%)
θ	torsional vibration amplitudes (rad)

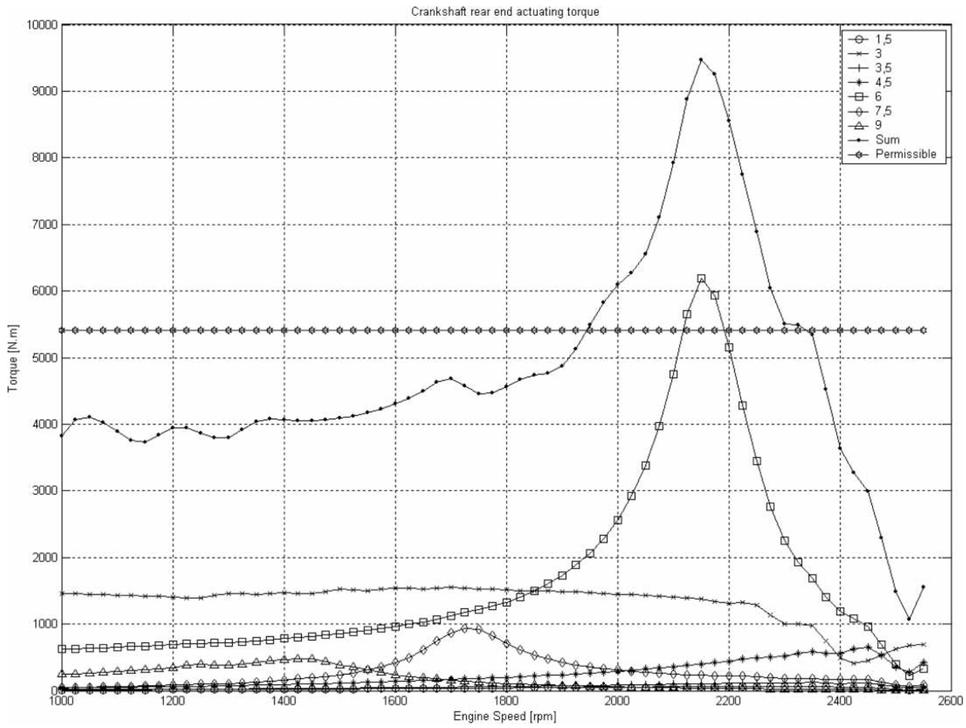

Fig. 31 Torque between the flywheel and the 6th cylinder without TVD

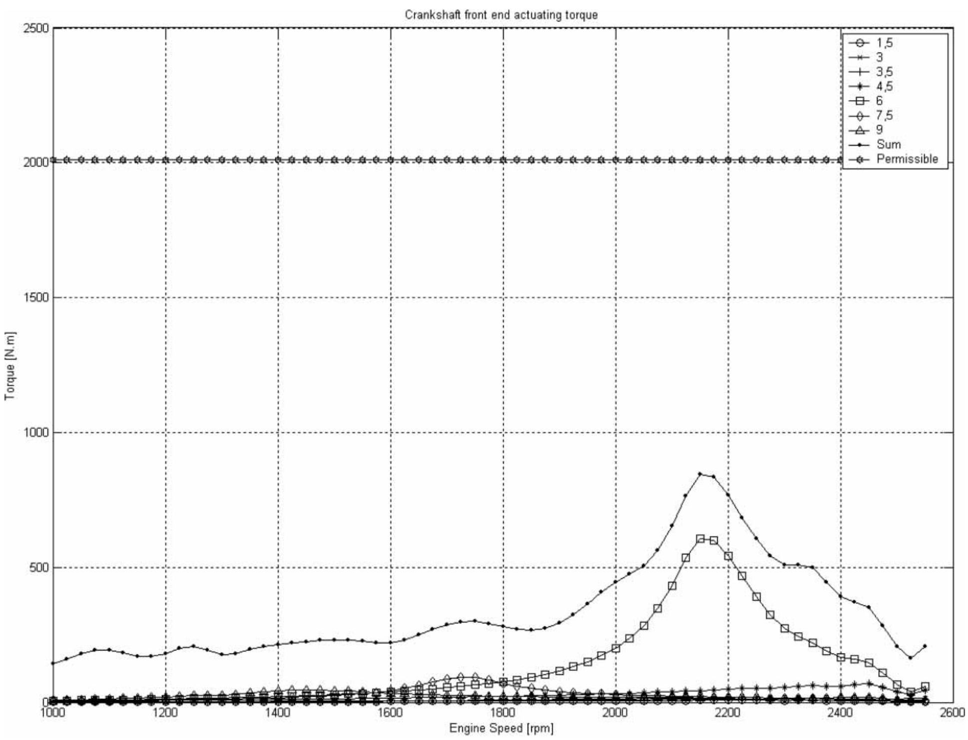

Fig. 32 Torque between the crankshaft pulley and the gear train without TVD

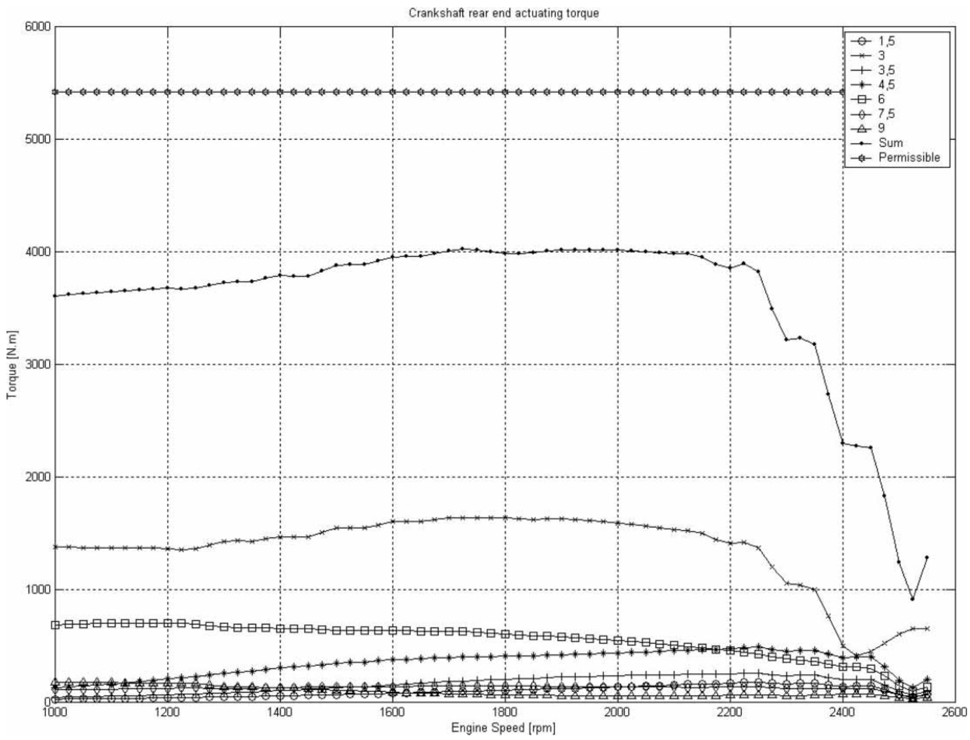

Fig. 33 Torque between the flywheel and the 6th cylinder with viscous TVD

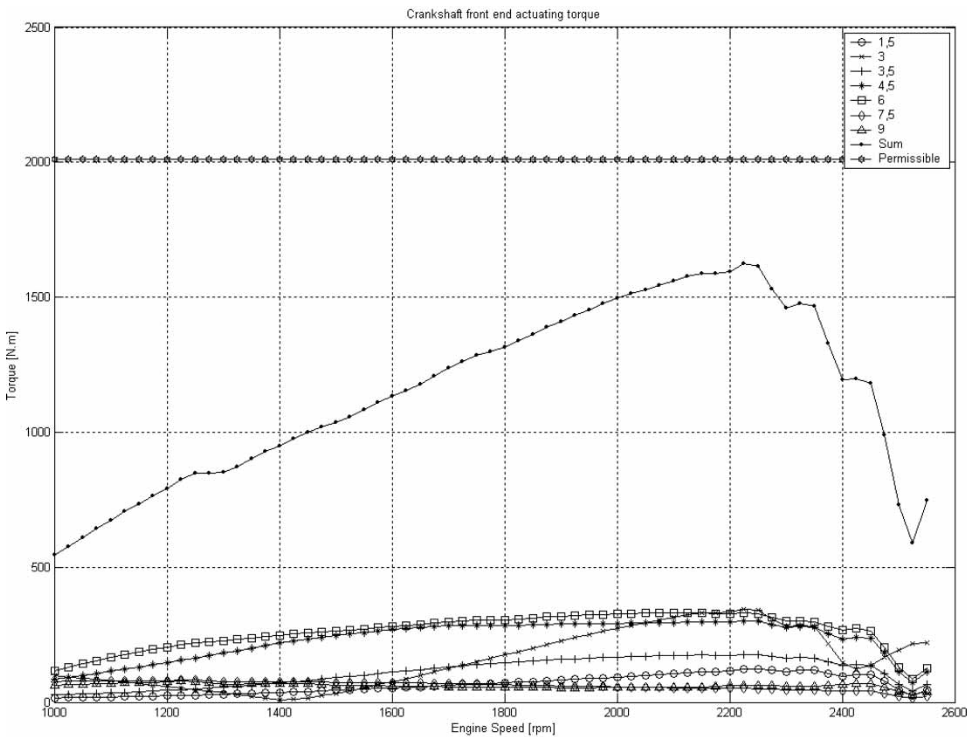

Fig. 34 Torque between the crankshaft pulley and the gear train with viscous TVD

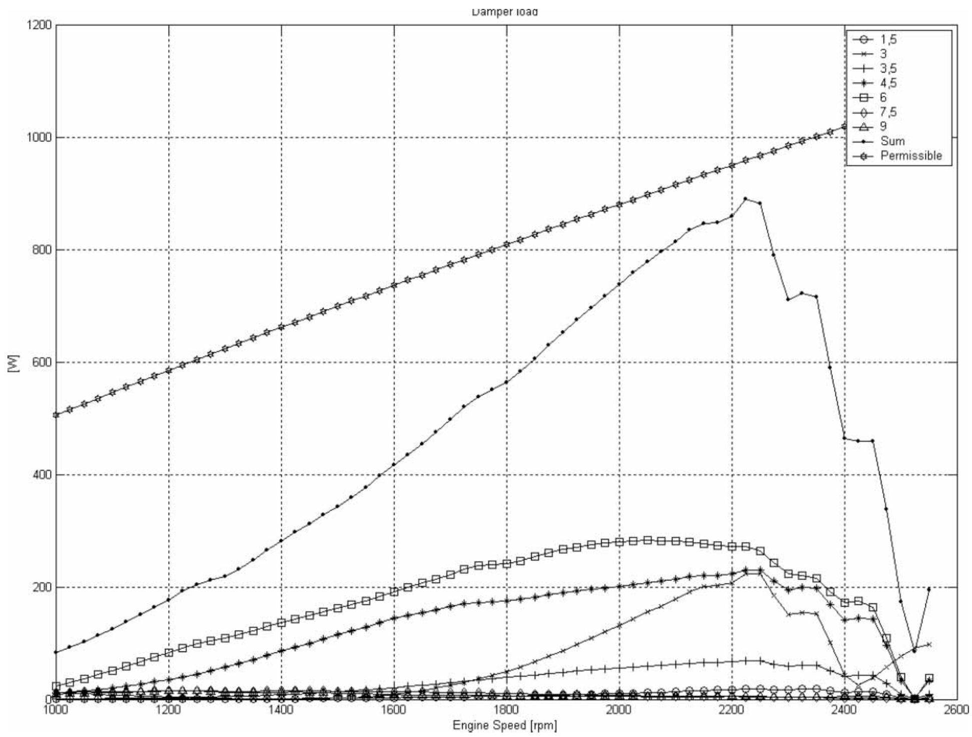

Fig. 35 Viscous damper load

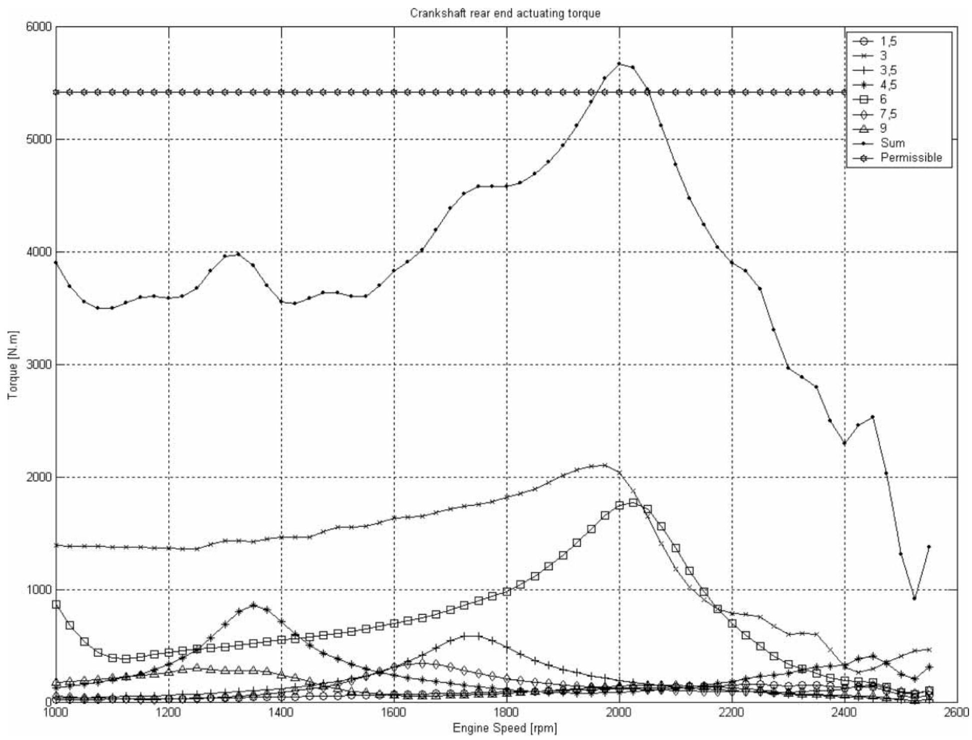

Fig. 36 Torque between the flywheel and the 6th cylinder with rubber TVD

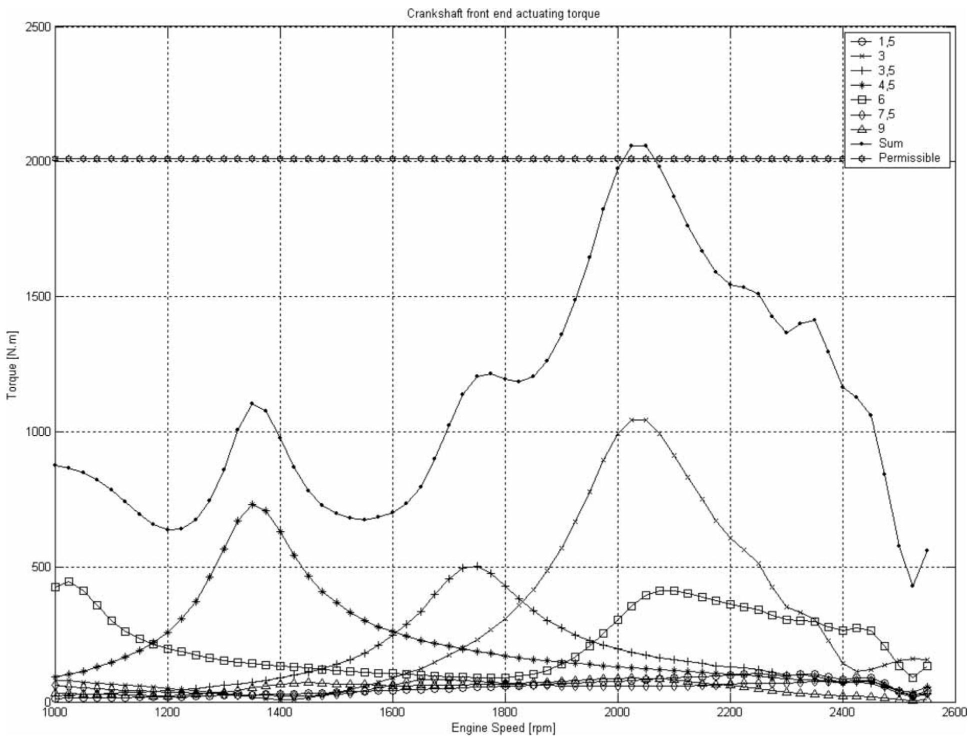

Fig. 37 Torque between the crankshaft pulley and the gear train with rubber TVD

λ	relation between crank radius and con rod length (–)
ν	kinematic viscosity (m^2/s)
τ	rubber shear stress (MPa)
ϕ	phase angle (rad)
$\mathbf{\Phi}$	transition state matrix (–)
χ	loss number (–)
ω	crankshaft angular velocity (rad/s)
ω_n	system natural frequency (rad/s)

APPENDIX 2

Torsional vibrations results

The graph in Fig. 31 depicts the dynamic torque between the flywheel and crankshaft connection versus the engine speed for the system without TVD. As can be observed, there is a maximum torque of 9500 Nm around 2200 r/min. If no dynamic response is considered, this value would be close to 3500 Nm, as indicated in Fig. 15. This is one of the aspects which indicate the importance of the TVA in the crankshaft's structural dimensioning.

The dynamic torque at the crankshaft pulley connection can be evaluated likewise. Figure 32 shows these results. In this case, the dynamic torque is much lower than the permissible one since there is no TVD assembled at this position.

The permissible torque was calculated considering the geometric dimensions of the crankshaft ends and the minimum bolt tightening forces.

The actuating torque at the rear end of the crankshaft, considering a viscous TVD, was reduced more than twofold, thus rendering this engine suitable for heavy-duty applications (Fig. 33).

Figure 34 shows the dynamic torque at the front end of the crankshaft. Compared to the system without TVD, this torque increased considerably in response to the influence of the absorber.

Figure 35 illustrates the dissipated power at the viscous TVD considering all the orders of Fourier series and the permissible damper load. The maximum value occurs close to 2200 r/min and, considering the absorber heat dissipation capability, one can conclude that no overloading will occur.

Figures 36 and 37 show the dynamic torque for the system with the rubber TVD at the rear and front ends, respectively. As can be seen, both regions present some overloading close to 2000 r/min.

APPENDIX 3

Summary of results

(1) Maximum amplitudes:

Without TVD – main orders	6/I	7.5/I	9/I
Amplitude (degrees)			
Calculated	1.12	0.17	0.09
Measured	1.17	0.17	0.08
Viscous TVD – main orders	3/I	4.5/I	6/I

Amplitude (degrees)			
Calculated	0.17	0.14	0.11
Measured	0.18	0.15	0.10

Rubber TVD – main orders	3/I	4.5/I	6/I	6/II
Amplitude (degrees)				
Calculated	0.33	0.24	0.14	0.33
Measured	0.32	0.25	0.12	0.32

(2) Maximum torques at crankshaft ends:

TVD	Without	Viscous	Rubber	Permissible
Torque (Nm)				
Front-end	850	1593	2070	2012
Rear-end	9484	4011	5673	5413

(3) Generated power at TVD:

TVD type	Damper load (W)	Permissible (W)
Viscous	882	960
Rubber (W)		
1st ring	1132	256
2nd ring	322	240

(4) Rubber shear stress:

Shear stress (MPa)	Calculated	Permissible
Stress		
1st ring	0.56	0.30
2nd ring	0.32	0.30